

CHEMISTRY BOARD QUESTION PAPER: JULY 2024

Time: 3 Hrs. Max. Marks: 70

General Instructions:

1. The question paper is divided into four sections.
2. Section A: Q.1 – Ten MCQs of 1 mark each. Q.2 – Eight very short answers of 1 mark each.
3. Section B: Q.3 to Q.14 – Twelve questions of 2 marks. Attempt any eight.
4. Section C: Q.15 to Q.26 – Twelve questions of 3 marks. Attempt any eight.
5. Section D: Q.27 to Q.31 – Five questions of 4 marks. Attempt any three.
6. Use of log table is allowed. Calculator not allowed.
7. Figures to the right indicate full marks.
8. For MCQs, write the correct answer with alphabet.

Constants: $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, $N_A = 6.022 \times 10^{23}$, $F = 96500 \text{ C}$

SECTION – A

Q.1. Select and write the correct answers: [10]

i. Integrated rate law for zero-order reaction is –

- (A)
- $kt = [A]_0 - [A]_t$
- (B)
- $k = 2.303/t \log([A]_0/[A]_t)$
- (C)
- $k = ([A]_0 - [A]_t)/t$
- (D)
- $2.303 \log[A]_0$

ii. Spin-only magnetic moment of Fe^{2+} –

- (A) 3.806 BM (B) 4.899 BM (C) 5.796 BM (D) 6.817 BM

iii. Relation between r and a in simple cubic –

- (A)
- $\sqrt{3}r = 4a$
- (B)
- $\sqrt{3}a = 4r$
- (C)
- $r = a/2$
- (D)
- $r = a/\sqrt{2}$

iv. Monomer used for thermocol

- (A) vinyl chloride (B) acrylamide (C) butadiene–acrylonitrile (D) styrene

v. pH of 0.01 M HCl –

- (A) 1 (B) 2 (C) 3 (D) 4

vi. α -amino acid without chiral centre –

- (A) alanine (B) valine (C) leucine (D) glycine

vii. Benzylic halide –



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viii. Neutral complex –

(A) $[\text{Co}(\text{NO}_2)_3(\text{NH}_3)_3]$ (B) $\text{Na}_3[\text{Co}(\text{NO}_2)_6]$ (C) $[\text{Co}(\text{NH}_3)_5\text{CO}_3]\text{Cl}$ (D) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$

ix. Reaction converting toluene to benzaldehyde –

(A) Etard (B) Gattermann–Koch (C) Friedel–Crafts (D) Swarts

x. Best reagent to convert primary alcohol $\rightarrow$ aldehyde –

(A) CrO_3 (B) PCC (C) KMnO_4 (D) $\text{K}_2\text{Cr}_2\text{O}_7$

Q.2. Answer the following: [8]

i. Name insecticide used instead of DDT.

ii. Hydrocarbon formed when CH_3MgI reacts with NH_3 .

iii. Calculate E_{cell} for $\text{Zn}|\text{Zn}^{2+}(0.08\text{M})||\text{Cr}^{3+}(0.1\text{M})|\text{Cr}$ using $E^\circ_{\text{Zn}} = -0.76\text{ V}$, $E^\circ_{\text{Cr}} = -0.74\text{ V}$.

iv. IUPAC name of $\text{Na}_3[\text{AlF}_6]$.

v. Electronic configuration of Zn^{2+} ($Z = 30$).

vi. IUPAC name of amine from Hofmann bromamide of acetamide.

vii. Condition for spontaneity with respect to entropy.

viii. Relation between van't Hoff factor (i) and degree of dissociation (α).

SECTION – B

Attempt any EIGHT [16]

Q.3. Boiling point elevation: ethyl acetate bp 77.06°C ; solution of 50 g solute in 150 g ethyl acetate boils at 82.6°C . $K_b = 2.77^\circ\text{C kg mol}^{-1}$. Calculate molar mass.

Q.4. Deduce first law for (a) isothermal process (b) isochoric process.

Q.5. Complete: (i) $\text{XeF}_2 + \text{H}_2\text{O} \rightarrow ?$ (ii) $\text{Xe} + \text{F}_2 \rightarrow ?$

Q.6. Differentiate between ionic and covalent crystals.

Q.7. Calculate mass of Cu deposited when 5 A current flows for 100 minutes. $M = 63.5\text{ g mol}^{-1}$.

Q.8. Define buffer solution. Write Henderson–Hasselbalch equation for acidic buffer.

Q.9. Write any two principles of green chemistry.

Q.10. Write preparation of Nylon–6 and two uses.

Q.11. Define: (a) Pyrometallurgy (b) Roasting

Q.12. Write salient features of SN_1 mechanism.

Q.13. What is denaturation of proteins? Name one pyrimidine base in RNA.

Q.14. Write reactions when ethylamine is treated with (a) excess ethyl bromide (b) Hinsberg reagent.

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SECTION – C**Attempt any EIGHT [24]**

- Q.15. Construct cell using SHE and Zn electrode. Write reaction and representation.
Calculate E_{cell} for 0.01 M Zn^{2+} using $E^\circ = +0.76\text{V}$.
- Q.16. State Henry's law. Write four conditions for ideal solutions.
- Q.17. Give one example each of ionisation, linkage and hydrated isomerism.
- Q.18. Calculate ΔH° for: $\text{SiO}_2 + 3\text{C} \rightarrow \text{SiC} + 2\text{CO}$ using given thermochemical equations.
- Q.19. Derive Ostwald's dilution law. Derive relation between K_{sp} and solubility of $\text{Al}(\text{OH})_3$.
- Q.20. 60% of reactant decomposes in 45 min (1st order). Calculate half-life. Write relation between $t_{1/2}$ and initial concentration for zero-order.
- Q.21. Write conditions for colour of transition metal ions. Write two uses of neon.
- Q.22. Explain anomalous behaviour of oxygen (atomicity, magnetic property, oxidation state).
- Q.23. Explain Cross Cannizzaro reaction. Draw Zwitter ion of sulfanilic acid.
- Q.24. Identify A, B, C in $\text{CH}_3\text{--CH}_2\text{Cl} + \text{KOH}(\text{aq}) \rightarrow \text{A} \xrightarrow{(\Delta, \text{Cu})} \text{B} \xrightarrow{(\text{powder})} \text{C}$. Rewrite reactions.
- Q.25. Explain optical isomerism in 2-chlorobutane.
- Q.26. Write Clemmensen reduction of propanone. Write Fehling's test for ethanol.

SECTION – D**Attempt any THREE [12]**

- Q.27. How is O_2 prepared from KClO_3 ? Write reactions for: (a) phenol $\rightarrow$ benzoquinone
(b) phenol $\rightarrow$ cyclohexanol
- Q.28. Define standard enthalpy of combustion and enthalpy of atomisation. Draw structure of D-ribose. Write uses of bronze alloy.
- Q.29. Distinguish between order and molecularity. Write gases liberated at anode and cathode during electrolysis of NaCl . Write two uses of neon.
- Q.30. Gold crystallises in fcc; edge = 408 pm. Calculate density ($M = 197 \text{ g mol}^{-1}$). Name one self-cleaning flower. Write monomer for PVC.
- Q.31. Explain formation of $[\text{Co}(\text{NH}_3)_6]^{3+}$ using VBT. Convert: (a) acetic acid $\rightarrow$ ethyl acetate
(b) acetic acid $\rightarrow$ ethyl alcohol.

